Everett D. McArthur

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Summary

Black Hole Scaling Relationships, Galactic Dynamics and Galactic Archeology through simulations and observations.

Education

The Ohio State University Aug 2024 - Present

PhD (Astronomy)

Columbia University Sep 2019 - May 2023

BA (Astronomy)

Awards and Honors

Whitten Andean and Amazonian Studies Scholarship Nov 2026
1st Place, 2024 Black Space Week Graduate Student Research Competition Jun 2024
Keynote Speaker, Passaic Arts and Science Charter School High School Commencement Jun 2023
Keynote Speaker, iLearn Schools: Empowerment in Teaching Jun 2023
Team-Up Together Scholarship Nov 2022
Robert R. Brookhart Scholarship Nov 2022

Research Appointments

Graduate Research Assistant Aug 2024 - present

The Ohio State University
Mentor(s): Dr. Jennifer Johnson, Dr. Lucy Lu

Graduate Research Assistant Aug 2024 - present

The Ohio State University
Mentor(s): Dr. Ji Wang and Dr. Caprice Phillips

KIPAC Post-Baccalaureate Research Fellow Jun 2023 - Aug 2024

Stanford University: Kavli Institute for Particle Astrophysics and Cosmology
Mentor(s): Dr. Martin Millon, Dr. Meridith Powell, Dr. Risa Wechsler

Undergraduate Research Assistant Aug 2022 - Jun 2023

Department of Astronomy, American Museum of Natural History
Mentor(s): Dr. Jackie Faherty

NSF REU Intern Jun 2022 - Aug 2022

Department of Astronomy, American Museum of Natural History
Mentor(s): Dr. Johanna Vos and Dr. Jackie Faherty

Undergraduate Research Student Oct 2020 - Jan 2022

Department of Astronomy, Columbia University
Mentor(s): Douglas Grion Filho and Dr. Kathryn Johnston

Governor STEM Scholar Sep 2018 - May 2019

New Jersey Institute for Technology
Mentor(s): Kevin Pemberthy

Publications

McArthur, Everett (July 2026). "Quasars Acting as Strong Lenses Found in DESI DR1". In: *The Astrophysical Journal* 1006.2, p. 147. DOI: 10.3847/1538-4357/ae8014. URL: https://doi.org/10.3847/1538-4357/ae8014.

McArthur, Everett et al. 2026, "Retrieval Analysis of an L0 Brown Dwarf from KPIC," in prep.

McArthur, Everett et al. 2026, "Radial Migration in NIHAO-UHD," in prep.

McArthur, Everett and Smith, Paul et al. 2026, "JWST Gentle Giants: Retrieval Analysis of TOI 2031b," in prep.

Proposed Telescope Time

* Millon, Martin (July 2024). *Establishing quasar-host scaling relations with strong gravitational lensing.* HST Proposal. Cycle 32, ID. #17710.

* Gao, Peter (Mar. 2025). *The Warm Jupiter Opportunity for Understanding Giant Exoplanet Evolution.* JWST Proposal. Cycle 4, ID. #9025.

*Co-I.

Research Talks and Posters

Contributed Talk, Churning the Pot: Radial Migration in NIHAO-UHD, American Astronomical Society Winter Meeting — Jan 2027

Contributed Talk, Churning the Pot: Radial Migration in NIHAO-UHD, National Society of Black Physicists Conference (NSBP) — Nov 2026

Talk, Quasars As Strong Lenses Found in DESI DR2, US AirForce Academy — Oct 2026

Poster, JWST Gentle Giants: Retrieval Analysis of TOI 2031b, Exoplanet Atmospheres (AAS) — March 2026

Talk, Using Machine Learning to Find Quasars as Strong Lenses, KIPAC AI Workshop — May 2024

Contributed Talk, Quasars as Strong Lenses in DESI, National Society of Black Physicists Conference (NSBP) — Nov 2023

Poster, Stellar Flyys and Where to Find Them, Flatiron Institute, Center for Computational Astrophysics (Gotham Fest) — Jun 2023

Poster, Stellar Flybys and their Hidden Family, American Astronomical Society Winter Meeting (AAS) — Aug 2023

Contributed Talk, Stellar Flybys and Brown Dwarfs, National Society of Black Physicists Conference (NSBP) — Nov 2022

Poster, Stellar Flybys and Brown Dwarfs, Columbia University Astrofest — Dec 2022

Talk, Stellar Flybys and Brown Dwarfs, Department of Astronomy, American Museum of Natural History — Aug 2022

Talk, Stellar Flybys in Gaia DR3, Center for Computational Astrophysics (CCA) - Gaia Fête — Jun 2022

Poster, American Astronomical Society Winter Meeting (AAS, canceled due to COVID-19) — Jan 2021

Talk, Kean University — May 2019

Research Mentoring

Christian Geiman, Single Epoch Black Hole Masses using PyQSOFit, Polaris Mentorship Program

SUMMER SCHOOLS

L'Space Mission Concept Academy, Lead Astrophysicist – Developed a mission proposal for a lunar lander to analyze regolith — Summer 2021

CMB-S4 Summer School, Analyzed ACT and SDSS data for high-resolution CMB experiments; developed skills in cross-correlation, power spectra, and filtering techniques — Aug 2021

National Society of Black Physicists Scholars Program — Summer 2021
Trained in Git, SciPy, and chi-squared analysis.

Professional Appointments

Head of Alumni Relations — Feb 2023 - Jun 2023
iLearn Schools - Founded and led the Alumni Association and set the mission and vision for the organization

Teaching Assistant III — Sep 2022 - Dec 2022
Department of Astronomy, Columbia University

Mathematics Expert — Jun 2021 - Aug 2021
The Classroom Door

Leadership Appointments

Chapter President, NSBP/NSHP Student Chapter	2026 - 2027
Chapter Secretary, NSBP/NSHP Student Chapter	2025 - 2026
Committee Member, Governor's STEM Scholars Advisory	2022 - 2024
Events Chair, Scientists and Engineers Builders for a Better Society	2022
Executive Board, Columbia Science Review	2021

Seminars

Pathway to College and Beyond, Bergen Arts and Science Charter School — Dec 2025

Quasars as Strong Lenses in DESI DR1, University of Illinois Urbana-Champaign, Department of Astronomy, Journal Club — Feb 2026

Quasars as Strong Lenses in DESI DR1, University of Kansas, Department of Physics Seminar — Dec 2025

Quasars as Strong Lenses, Stanford University, Department of Physics, KIPAC Tea — Jun 2023

How to Create a Resume, Montclaire University, Career and Readiness Club — Mar 2022

Outreach

Presenter, Arne Slettabak Planetarium — 2024–Present
Community Representative, San Francisco Museum of African Diaspora — 2026
Guest Speaker, Bexley Library Summer Talk Series — 2024
Telescope Viewing, Telescope Operator/Speaker (Monthly Event at KIPAC) — 2023–2024
Astro on Tap, Fly-Fly Stars (Monthly Neighborhood Bar Talks) — 2022
Guest Speaker, iLearn Schools 15th Anniversary Gala — 2022
Guest Speaker, New Jersey Public Charter Schools Association Gala — 2022
Panelist, New York Public Charter Schools Association Conference — 2022
Columbia Blueshift (Columbia Astronomy Club), Intro to Stellar Flybys — 2022
Garfield Public Library, Public Observing Night — 2022
Garfield Public Library, Habitable Worlds — 2021
Panelist, New York Public Charter Schools Association Conference — 2021
Columbia Educational Outreach, Guest Lecturer — 2020
Frontiers of Science Ambassador, Intro Science Course for Undergraduates (Tutor) — 2020

Selected Press

AI Just Found Seven Rare Quasar Lenses Hidden Among 800,000 Cosmic Objects, and Astronomers Didn't Expect It AI (Link)
Ohio State astronomy researchers discover several super-bright quasar lenses (Link)
How do supermassive black holes grow? AI finds 7 spacetime-warping 'quasars' that could help solve the mystery (Link)
Seven possible quasar lenses emerge from AI scan of 800,000 DESI objects (Link)
Astronomers Discovers Super Bright Quasar Lenses (Link)
Universe Today Machine Learning Discovers Quasars as Strong Lenses (Link)
Using Machine Learning to Find Quasar Lenses in DESI Data (Link)
Phys.Org Machine Learning finds Quasars as Strong Lenses (Link)
The Clifton Times, iLearn Schools "Make It Happen!" Day (Link)
BronxNet, Open – The Bronx Arts and Science Charter School
BronxNet, Bronx Arts and Science Charter School Ribbon Cutting

PROFESSIONAL MEMBERSHIPS

National Society of Black Physicists — Nov 2020–Present
American Astronomical Society — Dec 2020–Present
American Physical Society — Nov 2022–Present

Science Collaborations

LSST Strong Lensing Science Collaboration — Sep 2023–Present
DESI Science Collaboration — Sep 2023–Present

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